

3.38.

$$P_1. \dot{X} = \begin{pmatrix} 6 & -3 \\ 2 & 1 \end{pmatrix} X$$

$$A = \begin{pmatrix} 6 & -3 \\ 2 & 1 \end{pmatrix}$$

$$P(\lambda) = \det(A - \lambda I) = \det \begin{pmatrix} 6-\lambda & -3 \\ 2 & 1-\lambda \end{pmatrix}$$

$$= (6-\lambda)(1-\lambda) + 6$$

$$= \lambda^2 - 7\lambda + 12$$

$$= (\lambda-3)(\lambda-4) = 0$$

$$\Rightarrow \lambda_1 = 3, \lambda_2 = 4.$$

i) $\lambda_1 = 3$

$$(A - 3I)V = 0 \Rightarrow \begin{bmatrix} 3 & -3 \\ 2 & -2 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$3V_1 - 3V_2 = 0, 2V_1 - 2V_2 = 0$$

$$\Rightarrow V_1 = V_2$$

$$\Rightarrow V = \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} V_1 \\ V_1 \end{bmatrix} = V_1 \begin{bmatrix} 1 \\ 1 \end{bmatrix} = C \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\Rightarrow X^1(t) = e^{\lambda_1 t} V = e^{3t} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \text{ (setting } C=1)$$

ii) $\lambda_2 = 4$

$$(A - 4I)V = 0 \Rightarrow \begin{bmatrix} 2 & -3 \\ 2 & -3 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = 0$$

$$\Rightarrow 2V_1 - 3V_2 = 0 \Rightarrow V_1 = \frac{3}{2}V_2$$

$$\Rightarrow V = \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} \frac{3}{2}V_2 \\ V_2 \end{bmatrix} = V_2 \begin{bmatrix} \frac{3}{2} \\ 1 \end{bmatrix} = C \begin{bmatrix} 3 \\ 2 \end{bmatrix}$$

$$\Rightarrow X^2(t) = e^{\lambda_2 t} V = e^{4t} \begin{bmatrix} 3 \\ 2 \end{bmatrix} \text{ (setting } C=1)$$

Thus, $X(t) = C_1 X^1(t) + C_2 X^2(t)$

$$= C_1 e^{3t} \begin{bmatrix} 1 \\ 1 \end{bmatrix} + C_2 e^{4t} \begin{bmatrix} 3 \\ 2 \end{bmatrix}$$

$$= \begin{bmatrix} C_1 e^{3t} + 3C_2 e^{4t} \\ C_1 e^{3t} + 2C_2 e^{4t} \end{bmatrix}$$

$$P_3. A = \begin{pmatrix} 3 & 2 & 4 \\ 2 & 0 & 2 \\ 4 & 2 & 3 \end{pmatrix}$$

$$P(\lambda) = \det(A - \lambda I) = \det \begin{bmatrix} 3-\lambda & 2 & 4 \\ 2 & -\lambda & 2 \\ 4 & 2 & 3-\lambda \end{bmatrix}$$

$$= -(\lambda-8)(\lambda+1)^2 = 0$$

$$\Rightarrow \lambda_1 = 8, \lambda_2 = \lambda_3 = -1 \text{ (double root)}$$

i) $\lambda_1 = 8$

$$(A - 8I)V = 0 \Rightarrow \begin{bmatrix} -5 & 2 & 4 \\ 2 & -8 & 2 \\ 4 & 2 & -5 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{cases} -5V_1 + 2V_2 + 4V_3 = 0 \\ 2V_1 - 8V_2 + 2V_3 = 0 \\ 4V_1 + 2V_2 - 5V_3 = 0 \end{cases} \Rightarrow \begin{cases} V_1 = V_3 \\ V_2 = \frac{V_3}{2} \end{cases}$$

$$\Rightarrow V = \begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix} = \begin{bmatrix} V_3 \\ \frac{V_3}{2} \\ V_3 \end{bmatrix} = V_3 \begin{bmatrix} 1 \\ \frac{1}{2} \\ 1 \end{bmatrix} = C \begin{bmatrix} 2 \\ 1 \\ 2 \end{bmatrix}$$

$$\Rightarrow X^1(t) = e^{\lambda_1 t} V = e^{8t} \begin{bmatrix} 2 \\ 1 \\ 2 \end{bmatrix} \text{ (} C=1)$$

ii) $\lambda_2 = \lambda_3 = -1$

$$(A + I)V = 0 \Rightarrow \begin{bmatrix} 4 & 2 & 4 \\ 2 & 1 & 2 \\ 4 & 2 & 4 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{cases} 4V_1 + 2V_2 + 4V_3 = 0 \\ 2V_1 + V_2 + 2V_3 = 0 \\ 4V_1 + 2V_2 + 4V_3 = 0 \end{cases} \Rightarrow V_2 = -2V_1 - 2V_3$$

$$\Rightarrow V = \begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix} = \begin{bmatrix} V_1 \\ -2V_1 - 2V_3 \\ V_3 \end{bmatrix} = \begin{bmatrix} V_1 \\ -2V_1 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ -2V_3 \\ V_3 \end{bmatrix}$$

$$= V_1 \begin{bmatrix} 1 \\ -2 \\ 0 \end{bmatrix} + V_3 \begin{bmatrix} 0 \\ -2 \\ 1 \end{bmatrix}$$

$$= C_1 \begin{bmatrix} 1 \\ -2 \\ 0 \end{bmatrix} + C_2 \begin{bmatrix} 0 \\ -2 \\ 1 \end{bmatrix}$$

$$\Rightarrow X^2(t) = e^{\lambda_2 t} \begin{bmatrix} 1 \\ -2 \\ 0 \end{bmatrix} = e^{-t} \begin{bmatrix} 1 \\ -2 \\ 0 \end{bmatrix}$$

$$X^3(t) = e^{\lambda_3 t} \begin{bmatrix} 0 \\ -2 \\ 1 \end{bmatrix} = e^{-t} \begin{bmatrix} 0 \\ -2 \\ 1 \end{bmatrix}$$

Thus $X(t) = C_1 X^1(t) + C_2 X^2(t) + C_3 X^3(t)$

$$= C_1 e^{8t} \begin{bmatrix} 2 \\ 1 \\ 2 \end{bmatrix} + C_2 e^{-t} \begin{bmatrix} 1 \\ -2 \\ 0 \end{bmatrix} + C_3 e^{-t} \begin{bmatrix} 0 \\ -2 \\ 1 \end{bmatrix}$$

$$P5. A = \begin{bmatrix} -7 & 0 & 6 \\ 0 & 5 & 0 \\ 6 & 0 & 2 \end{bmatrix}$$

$$P(\lambda) = \det(A - \lambda I) = \det \begin{bmatrix} -7-\lambda & 0 & 6 \\ 0 & 5-\lambda & 0 \\ 6 & 0 & 2-\lambda \end{bmatrix}$$

$$= -(5-\lambda)^2(\lambda+10) = 0$$

$$\Rightarrow \lambda_1 = -10, \lambda_2 = \lambda_3 = 5. \text{ (double root)}$$

$$i) \lambda_1 = -10.$$

$$(A + 10I)V = 0 \Rightarrow \begin{bmatrix} 3 & 0 & 6 \\ 0 & 15 & 0 \\ 6 & 0 & 12 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{aligned} 3v_1 + 6v_3 &= 0, & 15v_2 &= 0, & 6v_1 + 12v_3 &= 0 \\ \downarrow & & \downarrow & & & \\ v_1 &= -2v_3 & v_2 &= 0 & & \end{aligned}$$

$$\Rightarrow V = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = \begin{bmatrix} -2v_3 \\ 0 \\ v_3 \end{bmatrix} = v_3 \begin{bmatrix} -2 \\ 0 \\ 1 \end{bmatrix} = C \begin{bmatrix} -2 \\ 0 \\ 1 \end{bmatrix}$$

$$\Rightarrow x^1(t) = e^{\lambda_1 t} V = e^{-10t} \begin{bmatrix} -2 \\ 0 \\ 1 \end{bmatrix}$$

$$ii) \lambda_2 = \lambda_3 = 5.$$

$$(A - 5I)V = 0 \Rightarrow \begin{bmatrix} -12 & 0 & 6 \\ 0 & 0 & 0 \\ 6 & 0 & -3 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{aligned} -12v_1 + 6v_3 &= 0 \\ 6v_1 - 3v_3 &= 0 \end{aligned} \Rightarrow v_3 = 2v_1$$

$$\Rightarrow V = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = \begin{bmatrix} v_1 \\ v_2 \\ 2v_1 \end{bmatrix} = \begin{bmatrix} v_1 \\ 0 \\ 2v_1 \end{bmatrix} + \begin{bmatrix} 0 \\ v_2 \\ 0 \end{bmatrix}$$

$$= v_1 \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix} + v_2 \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

$$= C_1 \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix} + C_2 \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

$$\Rightarrow x^2(t) = e^{\lambda_2 t} \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix} = e^{5t} \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}$$

$$x^3(t) = e^{\lambda_3 t} \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = e^{5t} \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

$$\text{Thus, } x(t) = C_1 x^1(t) + C_2 x^2(t) + C_3 x^3(t)$$

$$= C_1 e^{-10t} \begin{bmatrix} -2 \\ 0 \\ 1 \end{bmatrix} + C_2 e^{5t} \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix} + C_3 e^{5t} \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

$$P7. A = \begin{pmatrix} 1 & 1 \\ 4 & 1 \end{pmatrix}, x(0) = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$$

$$P(\lambda) = \det(A - \lambda I) = \det \begin{pmatrix} 1-\lambda & 1 \\ 4 & 1-\lambda \end{pmatrix}$$

$$= \lambda^2 - 2\lambda - 3 = (\lambda - 3)(\lambda + 1) = 0$$

$$\Rightarrow \lambda_1 = 3, \lambda_2 = -1.$$

$$i) \lambda_1 = 3.$$

$$(A - 3I)V = 0 \Rightarrow \begin{bmatrix} -2 & 1 \\ 4 & -2 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow v_2 = 2v_1$$

$$\Rightarrow V = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} v_1 \\ 2v_1 \end{bmatrix} = v_1 \begin{bmatrix} 1 \\ 2 \end{bmatrix} = C \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

$$\Rightarrow x^1(t) = e^{3t} \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

$$ii) \lambda_2 = -1.$$

$$(A + I)V = 0 \Rightarrow \begin{bmatrix} 2 & 1 \\ 4 & 2 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow v_2 = -2v_1$$

$$\Rightarrow V = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} v_1 \\ -2v_1 \end{bmatrix} = v_1 \begin{bmatrix} 1 \\ -2 \end{bmatrix} = C \begin{bmatrix} 1 \\ -2 \end{bmatrix}$$

$$\Rightarrow x^2(t) = e^{-t} \begin{bmatrix} 1 \\ -2 \end{bmatrix}$$

$$\text{Thus, } x(t) = C_1 e^{3t} \begin{bmatrix} 1 \\ 2 \end{bmatrix} + C_2 e^{-t} \begin{bmatrix} 1 \\ -2 \end{bmatrix}$$

From the initial conditions, we get

$$x(0) = \begin{pmatrix} 2 \\ 3 \end{pmatrix} = \begin{pmatrix} C_1 + C_2 \\ 2C_1 - 2C_2 \end{pmatrix}$$

$$\Rightarrow \begin{cases} C_1 + C_2 = 2 \\ 2C_1 - 2C_2 = 3 \end{cases} \Rightarrow C_1 = \frac{7}{4}, C_2 = \frac{1}{4}$$

$$\Rightarrow x(t) = \frac{7}{4} e^{3t} \begin{bmatrix} 1 \\ 2 \end{bmatrix} + \frac{1}{4} e^{-t} \begin{bmatrix} 1 \\ -2 \end{bmatrix}$$

$$P8. A = \begin{bmatrix} 1 & -3 \\ -2 & 2 \end{bmatrix}, x(0) = \begin{pmatrix} 0 \\ 5 \end{pmatrix}$$

$$P(\lambda) = \det(A - \lambda I) = \det \begin{bmatrix} 1-\lambda & -3 \\ -2 & 2-\lambda \end{bmatrix}$$

$$= \lambda^2 - 3\lambda - 4 = (\lambda - 4)(\lambda + 1) = 0$$

$$\Rightarrow \lambda_1 = 4, \lambda_2 = -1$$

$$i) \lambda_1 = 4.$$

$$(A - 4I)V = 0 \Rightarrow \begin{bmatrix} -3 & -3 \\ -2 & -2 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow v_2 = -v_1$$

$$\Rightarrow V = \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} V_1 \\ -V_1 \end{bmatrix} = V_1 \begin{bmatrix} 1 \\ -1 \end{bmatrix} = C \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$\Rightarrow X^1(t) = e^{4t} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$ii) \lambda_2 = -1$$

$$(A+I)V=0 \Rightarrow \begin{bmatrix} 2 & -3 \\ -2 & 3 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow V_1 = \frac{3}{2} V_2$$

$$\Rightarrow V = \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} \frac{3}{2} V_2 \\ V_2 \end{bmatrix} = V_2 \begin{bmatrix} \frac{3}{2} \\ 1 \end{bmatrix} = C \begin{bmatrix} 3 \\ 2 \end{bmatrix}$$

$$\Rightarrow X^2(t) = e^{-t} \begin{bmatrix} 3 \\ 2 \end{bmatrix}, \text{ Thus, } X(t) = C_1 e^{4t} \begin{bmatrix} 1 \\ -1 \end{bmatrix} + C_2 e^{-t} \begin{bmatrix} 3 \\ 2 \end{bmatrix} \Rightarrow X^3(t) = e^{2t} \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

Using the initial conditions, we get

$$X(0) = \begin{pmatrix} 0 \\ 5 \end{pmatrix} = \begin{pmatrix} C_1 + 3C_2 \\ -C_1 + 2C_2 \end{pmatrix}$$

$$\Rightarrow \begin{cases} C_1 + 3C_2 = 0 \\ -C_1 + 2C_2 = 5 \end{cases} \Rightarrow \begin{cases} C_1 = -3 \\ C_2 = 1 \end{cases}$$

$$\Rightarrow X(t) = -3e^{4t} \begin{bmatrix} 1 \\ -1 \end{bmatrix} + e^{-t} \begin{bmatrix} 3 \\ 2 \end{bmatrix}$$

$$P.9. A = \begin{bmatrix} 3 & 1 & -1 \\ 1 & 3 & -1 \\ 3 & 3 & -1 \end{bmatrix}, X(0) = \begin{pmatrix} 1 \\ -2 \\ -1 \end{pmatrix}$$

$$P(\lambda) = \det(A - \lambda I) = \det \begin{bmatrix} 3-\lambda & 1 & -1 \\ 1 & 3-\lambda & -1 \\ 3 & 3 & -1-\lambda \end{bmatrix}$$

$$= -(\lambda-1)(\lambda-2)^2 = 0$$

$$\Rightarrow \lambda_1 = 1, \lambda_2 = \lambda_3 = 2. \text{ (double root)}$$

$$i) \lambda_1 = 1$$

$$(A-I)V=0 \Rightarrow \begin{bmatrix} 2 & 1 & -1 \\ 1 & 2 & -1 \\ 3 & 3 & -2 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow \begin{cases} 2V_1 + V_2 - V_3 = 0 \\ V_1 + 2V_2 - V_3 = 0 \end{cases} \Rightarrow V_1 - V_2 = 0 \Rightarrow V_1 = V_2$$

$$3V_1 + 3V_2 - 2V_3 = 0 \Rightarrow V_3 = 3V_2$$

$$\Rightarrow V = \begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix} = \begin{bmatrix} V_2 \\ V_2 \\ 3V_2 \end{bmatrix} = V_2 \begin{bmatrix} 1 \\ 1 \\ 3 \end{bmatrix} = C \begin{bmatrix} 1 \\ 1 \\ 3 \end{bmatrix}$$

$$\Rightarrow X^1(t) = e^t \begin{bmatrix} 1 \\ 1 \\ 3 \end{bmatrix}$$

$$ii) \lambda_2 = \lambda_3 = 2$$

$$(A-2I)V=0 \Rightarrow \begin{bmatrix} 1 & 1 & -1 \\ 1 & 1 & -1 \\ 3 & 3 & -3 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix} = 0$$

$$\Rightarrow V_3 = V_1 + V_2$$

$$\Rightarrow V = \begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix} = \begin{bmatrix} V_1 \\ V_2 \\ V_1 + V_2 \end{bmatrix} = \begin{bmatrix} V_1 \\ 0 \\ V_1 \end{bmatrix} + \begin{bmatrix} 0 \\ V_2 \\ V_2 \end{bmatrix}$$

$$= V_1 \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} + V_2 \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$$

$$= C_1 \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} + C_2 \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$$

$$X^3(t) = e^{2t} \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

Thus, the general solution is given by.

$$X(t) = C_1 X^1(t) + C_2 X^2(t) + C_3 X^3(t)$$

$$= C_1 e^t \begin{bmatrix} 1 \\ 1 \\ 3 \end{bmatrix} + C_2 e^{2t} \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} + C_3 e^{2t} \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$$

$$X(0) = \begin{bmatrix} 1 \\ -2 \\ -1 \end{bmatrix} \Rightarrow C_1 \begin{bmatrix} 1 \\ 1 \\ 3 \end{bmatrix} + C_2 \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} + C_3 \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ -2 \\ -1 \end{bmatrix}$$

$$\begin{cases} C_1 + C_2 = 1 \\ C_1 + C_3 = -2 \\ 3C_1 + C_2 + C_3 = -1 \end{cases} \Rightarrow \begin{cases} C_1 = 0 \\ C_2 = 1 \\ C_3 = -2 \end{cases}$$

Thus,

$$X(t) = e^{2t} \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} - 2e^{2t} \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$$

$$= e^{2t} \begin{bmatrix} 1 \\ -2 \\ -1 \end{bmatrix}$$